Secure wiring and finish the design

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Document permanent connections and evaluate added mass or friction.

SUCCESS CRITERIA - I can...

1. Check circuit continuity and mechanical security.
2. Preserve moving-part clearance during decoration.
3. Relate a design revision to an energy transfer or force.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “strain relief” and “insulation” to explain your answer. Sentence starters are optional.

strain relief: support that keeps a pull from stressing an electrical joint

insulation: material that limits unwanted electric conduction

short circuit: an unintended low-resistance path bypassing a component

01 Warm-up: bring back an earlier idea

At unchanged wheel rpm, circumference becomes **4** times as large. **Compare ideal travel speed** and **state one real-world limitation**.

Word help: circumference = distance around the edge of a circle

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: I = **0** with power disconnected.
Disconnect before soldering/securing wires. Use insulation and strain relief; route wires clear of gears. Follow supervised tool procedures.

Worked reasoning: A secure circuit also needs strain relief: a pull on a wire should not be carried solely by a solder joint. Insulation prevents exposed conductors from making unintended contacts.

Your turn: Draw one wire route showing a connection, insulation, strain relief and clearance from a rotating axle.

Sentence starter: In my work, I used … to show … . I labeled it … .



Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Draw your final wiring route. **Mark soldered joints**, insulation, strain relief and the switch location. **Name one possible wheel or** gear interference point.

Sentence starter: In my work, I used … to show … . I labeled it … .



Marzano check - circle: ① ② ③

04 Apply the physics

A small classroom car’s mass increases from **200 g** to **250 g** after decorations are added.
1. **Calculate the added mass**. **Write both total masses** in kilograms.
2. **Predict how acceleration changes** if net forward force stays the same.
3. **Explain why you need** to **check whether** the force really stays the same.

Sentence starter: I predict … . I used … to make this prediction because … .

Design	Mass (kg)	Net force (N)	Acceleration
Before	___	___	___
After	___	___	___

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

A wire rubs a wheel. **Explain the possible effect** on energy transfer and propose a repair before another powered test.

Sentence starter: The step I would change is … . I would change it because I found … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

At fixed current, move from r to **4** r from a long straight wire. Using B proportional to **1**/r, compare field strength. **Mark the distances** and field directions you compare.

Word help: current = rate of charge flow

Sentence starter: Both … . They differ in … because … .

Current out of page (dot). Add field arrows.

Marzano check - circle: 1 2 3

07 Exit check

Record one quality check your partner completed and one revision supported by that check.

Sentence starter: I notice … . This supports … because … .



Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

1. Create a power-off inspection checklist with three specific observations that would make you stop before powering the car.
2. **Explain the concern behind each**.

Sentence starter: I think … . I can point to …, which shows …, so … .



Marzano check - circle: 1 2 3